

# Kailo Coaching Hub Audit

## Product review & data-platform architecture

21 August 2026

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### Defining Product Goals

- **Team-native, coach-facing training platform.** The team is the unit; the coach is the primary user.
- **Stronger analytical tools and coach customization** than TrainingPeaks or FinalSurge — the analytics and the ability to tailor the surface are the differentiators, not feature parity.
- **A coach-determined path for athletes to interact** that *reduces* the need for athlete buy-in. The less work an athlete has to do, the better the product: participation should be near-passive by default, with the coach deciding how much the athlete is asked to do.
- **Highly customizable.** Coaches change the information they see, where it sits, and what it is called — the layout is theirs, not a template they work around.
- **Expandable and collapsible.** The system is iterative: a coach can run it as a roster and a workout planner and nothing else, or turn on every analytic layer, and move between those states at will. Some coaches will want all of it and some almost none; both are first-class users.

### Next Steps (ordered)

1. **Restructure the hub into the five-tab layout** of §1.4 — Today, Upcoming, Team, Recruiting (v2), Settings — with the athlete page reached through the Team tab (roster, results, or the PR / progression boards) rather than as its own tab.
  2. **Athlete objects with histories and attributes.** Each athlete carries a history (race and training paces over time) *and* a set of learned attributes — e.g. “performs better on hilly courses,” “degrades more in rain / heat.” These attributes are estimated from data, they change as more data arrives, and the set will grow. The data platform (§2) exists to populate and keep refreshing them.
  3. **Text-to-structured-workout parsing.** The planner grid must turn “2 mile wu, 5 tempo, 3 cd” into a structured workout that can be analyzed post hoc; the current parser misreads such inputs (see the Planner and Workout planner audits). It also needs a **templating layer** — “2 mi @ <threshold>” — where the placeholder resolves per athlete from the model we host; today only absolute values parse.
  4. **Weather upgrade.** Move the adjustment metric from dew point to wet-bulb globe temperature, and show conditions for four windows — this morning / afternoon, next morning / afternoon — to cover doubles.
  5. **Data & enrichment platform build-out** (§2): divest from Athletic.net; per-race time, weather, and hill enrichment feeding the learned attributes.
  6. **Recruiting** — a *v2 product*. Reserved in the tab structure but not built in v1.
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# 1 Product Audit

## 1.1 Two axes

The desktop hub is organized on two axes.

- **Vertical axis — thematic.** A left pane (top on narrow screens) of **five tier-0 objects**, each containing pages (Figure 1). The tier-0 objects are *Before Practice*, *Share with the Team*, *After Practice*, *The Roster*, and *Housekeeping*. Their labels are not clickable; the pages beneath them are. This axis is the spine of the audit below — every page gets an analysis, and every tier-0 object gets a general assessment.

| BEFORE PRACTICE | SHARE WITH THE TEAM | AFTER PRACTICE | THE ROSTER  | HOUSEKEEPING      |
|-----------------|---------------------|----------------|-------------|-------------------|
| Today           | Athlete day         | Session review | Week wall   | Operations        |
| Tasks + Links   | Team board          | Inbox          | Volume      | Data health       |
| Planner         | 1-on-1s             |                | Athlete     | How this works P7 |
| Workout planner |                     |                | Parameters  | Setup             |
| Pace cards      |                     |                | Results     |                   |
|                 |                     |                | Predictions |                   |

Figure 1: Vertical (thematic) axis as of August 21, 2026

- **Horizontal axis — temporal.** The coach's day, as a left-to-right flow. Each step is simply a **hyperlink** to one of the vertical-axis pages (Figure 2).

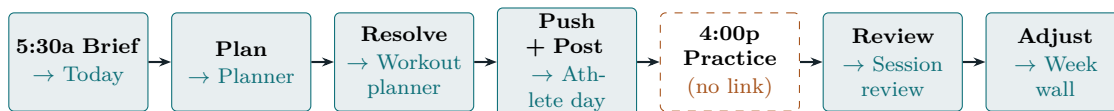


Figure 2: The horizontal (temporal) axis: the coach's day. Each step links to a vertical-axis page. **4:00p Practice currently links nowhere.**

**The day cuts across the thematic tiers.** The seven day-steps do not live inside one tier-0 object — they thread through four of them:

| Day step       | Links to        | Tier-0 home         |
|----------------|-----------------|---------------------|
| 5:30a Brief    | Today           | Before Practice     |
| Plan           | Planner         | Before Practice     |
| Resolve        | Workout planner | Before Practice     |
| Push + Post    | Athlete day     | Share with the Team |
| 4:00p Practice | — (nothing)     | —                   |
| Review         | Session review  | After Practice      |
| Adjust         | Week wall       | The Roster          |

[Future consideration should be made to decide whether information should be organized in a temporal or thematic axes.]

## 1.2 Vertical axis — page-by-page audit

## Before Practice

### Today [← day: 5:30a Brief]

This is the landing page ([kailo.fit/hub](#)) — a long stack of text widgets. Top widgets: “[N] uploaded yesterday,” “Available today,” “Data flags,” “Gone quiet,” “Need a look,” and “Dew point at practice.” Below them: “Where your attention goes” (activities flagged abnormal — average heart rate high on an easy run, distance longer than expected), a “Today’s plan” widget with a print option, and a practice conditions widget (“Conditions this afternoon”).

- **Keep high-quality, accessible print.** The plan’s print option is good and should stay easy to reach throughout.
- **Conditions is buried and single-window.** It sits too far down, does not currently populate, and only shows the afternoon — athletes running *doubles* lose the morning window entirely.
- **No expand/collapse** on any widget, producing a very long page.

**Works.** Everything on the page functions except the conditions widget, which is not populating — conditions were entered at 3 pm and the page says they update overnight, so this may resolve by morning.

**Assessment.** Too much information, almost all of it text, with no visual hierarchy to separate what matters. Because this is the landing page, deep customization is essential.

### Tasks + Links

This is a simple and considerably shorter page than the Today screen. It consists of two widgets: a basic task organizer and a place to store hyperlinks. The quick links widget prompts, “Add the ones you reach for weekly - entries, results, the conference site.”

**Works.** Both widgets function.

**Assessment.** A dedicated task page is likely useful, with a connected widget that can be added to a landing page. I don’t see much long-term need for the “Quick links” section. Anything that would go there should be put in its proper place.

### Planner [← day: Plan]

The planner page is a core aspect of the product. It has planning capabilities for the entire season (day, week, and entire training block). Weeks can be toggled one by one. There is a pin every Wednesday in this example. I am interested to know how the week grid works. Namely, does inputting text create a formatted workout? Ideally, one could write “Easy 45 minutes” or “2 mile wu, 5 tempo, 3 cd” in the grid and then recover an analysis of the workout based on how it was completed vs. predicted (easy pace was faster or higher-HR than normal; tempo pace is improving).

**Partial.** All 14 weeks arrive automatically assigned as base weeks; there is no way to designate a week as anything else up front — each can only be edited after the fact.

**Assessment.** This will be a core page going forward that should anchor a future “Upcoming” section. Technical considerations must be made to determine the degree to which text inputs can create structured workouts that can be analyzed post hoc.

### Workout planner [← day: Resolve]

This seems to be a more zoomed-in version of the planner day above. It allows for creating workout pods, prescribing paces for a workout, and making weather/terrain adjustments. Again, there is some instructional language here that is just a bit over the top, such as “The question: one date, six lanes. I plug the workout in, the groups and paces come out, weather and terrain already

applied.” I’m not sure what the “What moved since last week” section corresponds to, unless it’s simply product updates. There is a “Rules baked in” section, which is basically settings.

**Partial.** *The text input misparses, as noted; the dew-point adjuster does work.*

**Assessment.** Dew point is currently the listed weather metric used to determine weather adjustments. Based on the literature, we’d probably want this to move to wet-bulb globe temperature. The “Rules baked in” section should be removed and replaced with a link to the workout planner settings page, for organizational purposes. As above, we need to put in technical work to make the text-to-planner conversion work better. For instance, I tried to write “5 up, 2 tempo, 3 cd,” and the current iteration put “2 tempo, 3 cd” as the recovery.

## ■ Pace cards

Basically the VDOT training paces. It is unclear what “Provisional” means here. It asks users whether they want “page” or “phone” views. The prescribed paces are: “This week,” “Easy days,” “Long run,” “Weekly probe.”

**Works.** *Functions. The manual “page” / “phone” choice earns little, though: there is no mobile version, so a phone gets the desktop site either way.*

**Assessment.** This is again a core feature that should probably be linked in the training section but live under the Athlete pages. I don’t understand the logic for why certain paces are excluded — namely recovery and tempo/LT.

**Tier-0 assessment.** In general this tier is too much of a catchall for the core elements of the product. This isn’t necessarily a problem on its own. However, given how loaded this part is with features compared to the next two Tier-0 categories (Share with the Team and After Practice), it suggests there are better ways to organize these data and features to disperse them in a more coherent and balanced way. I would recommend creating a “Today” landing page and “Upcoming” planning sections that split up this data.

## Share with the Team

### ■ Athlete day [← day: Push + Post]

This page allows you to see what a given athlete sees on their Kailo app for each day’s practice. **Unverified.** *Not yet functionally tested.*

**Assessment.** This page can likely be a submodule of existing planners as a “Preview” option.

## ■ Team board

This page is a grab bag of different planning and team-related tools. “This week” is the first tab and is similar to the Workout planner. PR Board, currently unpopulated, seems to be a way to assemble PRs from a given meet (or overall PRs — it is unclear which). The team document, also currently unpopulated, seems to be a place to record team rules/creeds/values. Race plans allow the user to add different goals/plans for specific athletes for specific races. You can also input other teams’ results.

**Partial.** *Works well mechanically, but much of it is unformatted character strings — “8K” stays text rather than resolving to 8000m and populating event or pace — and a workout entered in “This week” does not update the Planner.*

**Assessment.** This page has too many disparate elements that limit its usefulness. While there is a conceptual core, there are better places for race info, weekly planning, team rules, and team PRs. A specific team homepage would be useful, which could contain important division/conference information along with the team document. Race plans can definitely be improved by adding both predictive modeling for a given athlete (predicted pace range) and MCMC simulations to see how that athlete stacks up against expected competition.

### 1-on-1s

This is a central repository to prepare and document 1-on-1s. The structure of the meeting notes isn't great since there is no expand/collapse, so long notes would render the page extremely long.

**Partial.** *Functions, but the question prompts are not editable, and notes do not collapse — a season of long meeting notes would be hard to move through.*

**Assessment.** I think this is probably a service that only some coaches will use, given that many won't input notes on meetings, but it is a good feature to have. However, like other features in this section, I think it could be moved to a better location.

**Tier-0 assessment.** Generally, the features in this tier are all individually useful, but some read as redundant with other features, and all of them could be placed in more intuitive locations. I would recommend removing this as an organizing tier and splitting its data among future Today, Upcoming, and Team sections.

## After Practice

### Session review [← day: Review]

Currently this page is not populated at all, which makes it hard for me to fully assess. But it seems to be information about the most recent session.

**Unverified.** *Does not appear to be working, but with nothing populated it cannot be verified either way.*

**Assessment.** This page is, of course, extremely central to coaching. We can create a “Most recent session results” module for the “Today” screen with the information in this section. But we need a space to put historical session data (e.g., the session from five days ago).

### Inbox

Some of these are repeated widgets from the current “Today” page (namely the “Needs a look” section, with some overlap in “Contact coverage”). The Reason ledger and Confounds sections are not fully populated for me, so I can't fully assess what's in them (though the latter seems to be some predictive or analytical system).

**Works.** *Functions. Linking email / communications with athletes would raise its value.*

**Assessment.** Similar to the Share with the Team section, I think there is important information in this section that would be better served by being moved under a different header in a more optimal organizing strategy.

**Tier-0 assessment.** I echo what I said in the Share with the Team section: many of these features are useful but are either redundant or not optimally placed. I also think that coaches need a way to toggle certain features on or off, such as meeting tracking with students. If a coach doesn't manually input that information, the app may nag them that they're never meeting with students. As part of the customization layer, it would be useful to treat some of the functionality like a grand-strategy video game: add layers/features as you want, or simply use it as a more basic tool of roster visuals and workout planning.

## The Roster

### Week wall [← day: Adjust]

This seems to be another calendar overview of the training week. For me, it is populated by only two athletes.

**Not working.** *Does not work in tandem with the Planner: tempo reps entered for Thursday render here as an easy run, not a workout.*

**Assessment.** Important information, but redundant with other pages. Perhaps other pages on training programs could have various view options, so you could have a “week view” option like this.

### Volume

A key page. Athlete name, 7-day and 28-day volume with wk/wk %, rest days, last upload, percent complete, directive, loop, and trend (in histogram form) all sit under the default “This window” pane. The second pane is “Plan & Goals,” which has weekly volume goals and season peak-mileage goals (and down-week depth). There are unpopulated Actuals and Adherence panes.

**Partial.** *Works, but the directive and loop fields never populate. Actuals and Adherence are empty, though that appears to be by design.*

**Assessment.** There needs to be a legend for certain terms or concepts such as Complete %, wk/wk %, directive, and loop. The trending inlay graph is pretty, but I'm not sure about its use. In general, lots of the information here is useful. But again, the Plan & Goals section would be better nestled in a planning section.

### Athlete

This is basically a roster. It is a drop-down. Upon selecting an athlete, you get “The person file” with high-school info, dietary restrictions, and class schedules. There is also information about surveys and check-ins, notes, and the athlete's stated goals. The page continues to Paperwork, Training room, Recruiting file, Class schedule, and Medical and emergency contact.

**Works.** *Every object on the page functions.*

**Assessment.** In general, this page is pretty long. This is more like an athlete's settings than a landing page. A more optimal landing page would be closer to a public roster: name, grade, gender, hometown, PRs, recent race results, and recent training (a week-over-week mileage graph, perhaps), along with some future-oriented things like goals. Dietary restrictions and class schedules could still be populated below, but some of the things currently on the page could simply be links to other sections instead of repeated widgets (such as 1-on-1 notes).

### Parameters

This is a list of the parameters produced through modeling for each athlete. These include critical speed, Rx width, D-prime, anchors, speed reserve, and extrapolation.

**Partial.** *Appears to work, but there is no visible way to update critical speed by hand, and speed reserve and D-prime are not populating.*

**Assessment.** The listed critical speed (with the CIs attached) could be an optional widget added to athlete profiles. This page, generally, is probably best kept in the settings.

## Results

As expected, a results page with tabs for “by meet,” “season progression,” “record board,” “courses,” “import,” “goals,” and “season review.”

**Works.** *Functions.*

**Assessment.** As said above, some of the recent results and PRs could probably be included as (optional) widgets on the athlete profile page. But a central results repository is still important to include in the “Team” section. The import option should be a hyperlink to a pop-up, not its own tab. Some of the remaining information could likely be either reorganized or broken into different pages.

## Predictions

This is a page devoted to assessing how accurate past race predictions were. Past results are shown (“Locked before the gun”) in a widget, followed by a grading widget, a calibration widget, and a larger season replay widget.

**Unverified.** *Currently empty, so it cannot be assessed.*

**Assessment.** I’m not sure how central this page is. Perhaps because no predictions have been made yet — and I can’t see current predictions — this page seems better nestled in the settings.

**Tier-0 assessment.** This is the section where not being able to click the tier-0 module name (The Roster) is most frustrating. I think the optimal flow would be a “Team” landing page with team name, picture, and conference/division information. This would then have a button for the roster, another for results, another for PRs, and perhaps season progression.

## Housekeeping

### Operations

This is a page for logistical/operational components of coaching, such as money or equipment information. It also allows .ics export of meet/training calendar information. The first two tabs — season board and this meet — seem to be repeats of existing data.

**Works.** *Functions.* “Paste receipts” should be upload receipts if the page is to serve auditing.

**Assessment.** This is probably better nestled under the Team label instead of the Housekeeping label, since I think the latter is better served as settings.

### Data health

This is more of a classic settings page on data importation. The first widget (Coverage) shows runs carrying a heart-rate average, session uploads carrying lap marks, and athletes with a band prescribed today. This is followed by a widget on whose data is not arriving. There is then a long widget of where each athlete’s data comes from (Strava vs. Garmin). There is also an extremely long section of flags from the data (“Things worth knowing about the data”). Finally, there is a widget showing who is missing paperwork.



**Partial.** Functions overall; two open items: whether the Strava linkage holds long-term, and athlete naming is inconsistent — some appear only as ids (`user-a5a41ffd`), some as a first name, some as first and last. The inconsistency is universal across the product.

**Assessment.** I’m not sure whether the paperwork described is medical clearances. If so, that information is too different from the data stuff here to be included. I think this could be recast as a “Data importation” settings menu.

### How this works

An unpopulated introduction/explainer for the app.

**Not working.** Completely empty.

**Assessment.** Obvious need.

### Setup

Lots of settings panels here, which can all be broken into their own settings tabs. These include “Recruit-visit mode,” which masks athlete names “so a campus visit sees the machinery and not the people.” There are also settings about program information, team information (name, location, coaches/access settings), and data flags. There are also bulk import and export widgets.

**Works.** Appears to work. Wish: an OSM-backed venue lookup — typing “Kezar Track” suggests “Kezar Track, San Francisco,” and clicking autopopulates the lat/long.

**Assessment.** These need to be built out as their own settings menu. This includes recruit-visit mode, team settings (including team/practice location), coach settings, bulk import/export, and the various other settings listed above.

**Tier-0 assessment.** There is a clear need to make this tier into a more standardized settings menu that centralizes many of the options scattered throughout the current system.

## 1.3 What works and what does not

Hands-on functional testing, 2026-08-23. Status key: **works** · **partial** · **not working** · **unverified**. The same notes appear inline on each page above.

**Cross-cutting findings** the table repeats in miniature:

- **Free text stays free text.** Across the product, what a coach types is stored as a character string — “8K” never becomes an event with a distance and a pace. This is the same gap as the text-to-structured-workout and templating work in the Next Steps.
- **Modules do not sync.** Team board’s “This week” and the Planner each hold their own copy of the week; the Planner and Week wall disagree about Thursday. One plan object, many views (§1.5) is the structural fix.
- **Athlete identity display is inconsistent everywhere** — ids, first names, and full names for different users on the same widgets.
- **There is no mobile version;** phones receive the desktop site.

## 1.4 Proposed structure: five main tabs

Collapse the sprawl into five tabs. This is a first pass, assembled from the per-page recommendations above — and because it goes to a group of coaches, the forward-looking pieces are put as questions (the “Q.” boxes) rather than decisions.



| Page                       | Status                             | Finding                                                                                                                                                                               |
|----------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Before Practice</i>     |                                    |                                                                                                                                                                                       |
| Today                      | <b>works</b>                       | only the conditions widget is not populating; conditions were entered at 3pm and the page says they update overnight — recheck in the morning                                         |
| Tasks + Links Planner      | <b>works</b><br><b>partial</b>     | —<br>all 14 weeks arrive automatically assigned as base; a week cannot be designated otherwise up front, only edited after the fact                                                   |
| Workout planner            | <b>partial</b>                     | text input misparses (as noted); the dew-point adjuster works                                                                                                                         |
| Pace cards                 | <b>works</b>                       | the manual page / phone toggle earns little: there is no mobile version, so a phone gets the desktop site                                                                             |
| <i>Share with the Team</i> |                                    |                                                                                                                                                                                       |
| Athlete day                | <b>unverified</b>                  | not yet functionally tested                                                                                                                                                           |
| Team board                 | <b>partial</b>                     | works well, but inputs stay unformatted strings (“8K” never resolves to 8000m / event / pace), and a workout entered in “This week” does not update the Planner                       |
| 1-on-1s                    | <b>partial</b>                     | question prompts are not editable; notes do not collapse, so a season of long notes is hard to move through                                                                           |
| <i>After Practice</i>      |                                    |                                                                                                                                                                                       |
| Session review             | <b>unverified</b>                  | appears non-functional, but nothing is populated, so it cannot be verified either way                                                                                                 |
| Inbox                      | <b>works</b>                       | wish: link email / communications with athletes                                                                                                                                       |
| <i>The Roster</i>          |                                    |                                                                                                                                                                                       |
| Week wall                  | <b>not working</b>                 | no tandem with the Planner: tempo reps entered for Thursday render here as an easy run                                                                                                |
| Volume                     | <b>partial</b>                     | works; directive and loop never populate (Actuals / Adherence empty by design)                                                                                                        |
| Athlete Parameters         | <b>works</b><br><b>partial</b>     | every object on the page functions<br>no visible way to hand-update critical speed; speed reserve and D-prime are not populating                                                      |
| Results Predictions        | <b>works</b><br><b>unverified</b>  | —<br>currently empty; cannot assess                                                                                                                                                   |
| <i>Housekeeping</i>        |                                    |                                                                                                                                                                                       |
| Operations                 | <b>works</b>                       | “paste receipts” should be <i>upload</i> receipts if the page is to serve auditing                                                                                                    |
| Data health                | <b>partial</b>                     | works; whether the Strava linkage holds long-term is unconfirmed, and athlete naming is inconsistent — some show as ids (user-a5a41ffd), some as a first name, some as first and last |
| How this works Setup       | <b>not working</b><br><b>works</b> | completely empty<br>wish: an OSM-backed venue lookup — typing “Kezar Track” suggests “Kezar Track, San Francisco” and clicking autopopulates the lat/long                             |

Table 1: Functional status by page, hands-on testing 2026-08-23.

1. **Today** — the landing page. Keeps what the current Today page gets right (the plan, with the print option easy to reach) and adds what the audit calls for: a “most recent session” module (from Session review), the needs-a-look flags, and practice conditions for *four* windows — this morning, this afternoon, next morning, next afternoon (fixing the doubles gap). Every widget collapsible, and the widget set coach-customizable.
2. **Upcoming** — planning, anchored by the Planner (day / week / training block) with the Workout planner as its zoomed-in day view, and more than one view of the same day (grid by group, sheet by athlete). Race plans (from Team board) are a v2 layer.
3. **Team** — a team landing page (name, picture, conference/division information, the team document), with routes into:
  - **Roster** — the public-roster-style list; selecting an athlete opens their page (layout below);
  - **Results** — the central results repository (by meet, season progression, record board, courses), where every athlete name links to that athlete’s page;
  - **PRs / season progression** — selecting an athlete goes to their page;
  - **Volume** — the volume tracking now in The Roster tier, plan & goals included;
  - **Operations** — logistics: money, equipment, the .ics calendar export.

The athlete page is deliberately *not* a top-level tab: the routes converge on it (Figure 7), so a coach reaches the same athlete object whether they come through the roster, the boards, or a result.

4. **Recruiting** — reserved as a tab, but a *v2 product*; not built in v1. (Recruit-visit mode — masking athlete names — ships in v1 under Settings.)
5. **Settings** — the standardized menu the Housekeeping audit calls for: data importation (Data health), team / coach / program settings, the workout planner rules, Parameters and Predictions, bulk import/export, recruit-visit mode, and the feature toggles (turn modules such as 1-on-1 tracking on or off).

Table 2 maps the proposal onto the current hub; the expanded plans below give the per-tab build detail.

**Q.** Does the five-tab split match how you actually move through a day — and is anything here missing a home?

## 1.5 Expanded design plans

### Today

The landing page answers “what do I need to know before practice” in one screen. Figure 3 is the *default* layout — every slot on it is a pick from the widget catalog below, so two coaches’ Today pages need not look alike.

- **Greeting line** — “Good morning, Coach. Today is Tuesday, September 8 · HCAC Preview in 26 days.” Time-of-day aware, with a deliberately simple default: the date plus days until the next meet. Deeper customization of this line (salutation, season day counts, or nothing at all) is a later conversation.
- **Glance row** — four small slots. The default four are uploaded yesterday, athletes available today, data flags / gone quiet, and needs a look; any slot can be swapped for another glance widget — days until the next race, paperwork outstanding, team mileage against plan (Table 3).

| Tab        | Routes / contents                                                                                                                          | Absorbs from current hub                                                                                              |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Today      | glance row; conditions (four windows); today's plan + print; needs a look; most recent session; tasks                                      | Today, Tasks + Links, Session review (as a module), Inbox (flags)                                                     |
| Upcoming   | the planner at four zooms: block → week → day → athlete preview; day views (grid / sheet); race plans (v2)                                 | Planner, Workout planner, Athlete day, Week wall (as a view), Team board (race plans, v2)                             |
| Team       | landing page; Roster; Results; PRs / progression; Volume; Operations; the athlete pages                                                    | The Roster tier (incl. Volume), Team board (team document, PR board), Operations, 1-on-1s (linked from athlete pages) |
| Recruiting | v2 placeholder                                                                                                                             | Recruiting file (from the person file)                                                                                |
| Settings   | data import; team / coach / program; workout rules; parameters; predictions; feature toggles; recruit-visit mode; bulk import/export; help | Data health, Setup, Parameters, Predictions, "Rules baked in," How this works                                         |

Table 2: The proposed five-tab structure, and where the current hub's pages land.

- **Conditions strip** — four windows (this morning, this afternoon, tomorrow morning, tomorrow afternoon) so doubles are covered. The adjustment model runs on wet-bulb globe temperature. The default display shows temperature, a conditions icon (sun / cloud / rain), precipitation chance, humidity, *and* the WBGT with the model's slowdown beside it — some coaches will want the number, some only the window's colour, so which weather fields appear is a coach preference.
- **Today's plan** — the print option kept one click away.
- **Most recent session** — the Session review material as a module, with historical sessions reachable from it.
- **Tasks** — the task organizer from Tasks + Links as a widget.

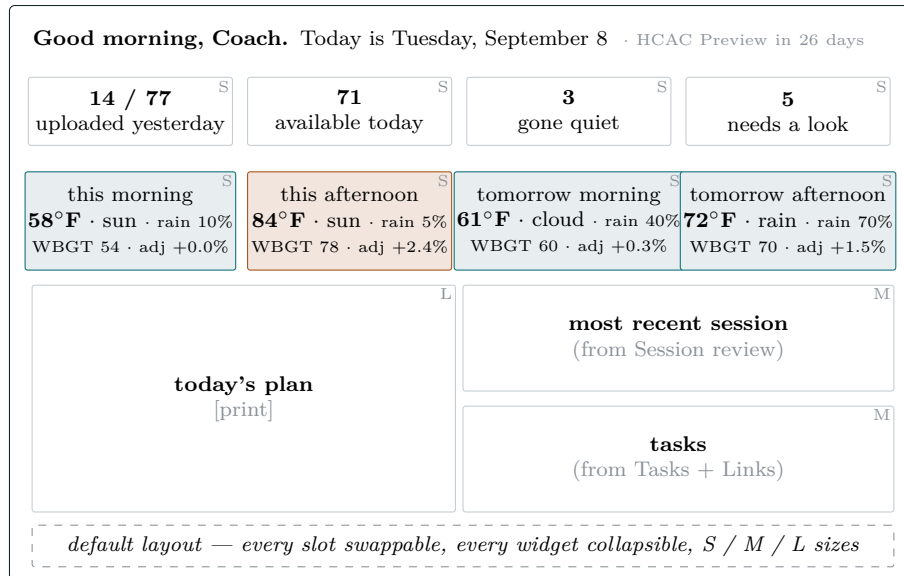


Figure 3: Today, default layout. The glance row is four swappable slots; each conditions window shows the readable fields (temperature, icon, rain) with the WBGT and the model’s slowdown beside them, and the WBGT band colours the window — the afternoon here is flagged. Which fields appear is a coach preference. Corner letters are widget sizes.

**Q.** Which four glance slots would you put in your row — and what is missing from the catalog below? For conditions: WBGT shown beside the slowdown, or only driving the window’s colour?

**Widgets come in three sizes.** Following the Apple widget model: a **glance** (S) widget is one number or flag in a 1×1 tile; a **medium** (M) widget is 2×1 and holds a short list or a compact chart; a **large** (L) widget is 2×2 and holds a real chart or a rich panel. The page is a grid the coach arranges — add, remove, reorder, and resize where a widget supports two sizes. Some widgets are text, some are numbers, some are charts; the catalog marks which, and the point of the chart widgets is that the coach never has to assemble a graph by hand.

Build order: the widget framework (grid, collapse, persistence) first, since everything else on the page is a widget; then the conditions strip; then the recent-session module; then the catalog grows one widget at a time, charts last.

## Upcoming

One planning object at four zoom levels, rather than three separate pages (Figure 5): the season block (phases, goals, peak mileage, down-week depth), the week grid (Week wall becomes a view option here, not a page), the single day (the Workout planner’s pods, paces, and weather/terrain adjustments), and the athlete preview (Athlete day as a “preview” button, not a destination).

- **Text-to-structured-workout parsing** is the load-bearing technical piece: “2 mile wu, 5 tempo, 3 cd” typed in the grid must become a structured workout that can be analyzed post hoc.
- **Workout templating has to be built.** The current system takes absolute values only. A workout must also be writable as a template — “2 mi @ <threshold>” — where the placeholder

| Glance widget (S)       | Shows                                                             | Type         |
|-------------------------|-------------------------------------------------------------------|--------------|
| Uploaded yesterday      | $n$ of $N$ athletes with a synced activity                        | count        |
| Available today         | athletes cleared to practice                                      | count        |
| Data flags / gone quiet | athletes with no upload in $k$ days                               | count, links |
| Needs a look            | activities flagged abnormal (HR high on easy, distance off plan)  | count, links |
| Days until next race    | countdown plus the meet name                                      | number       |
| Season day              | day count within the current season, or weeks to the championship | number       |
| Injured / training room | athletes on a modified plan                                       | count, links |
| Paperwork outstanding   | missing clearances                                                | count, links |
| Team mileage vs. plan   | this week's volume as a percent of planned                        | percent      |
| Check-ins missing       | surveys or check-ins not returned                                 | count        |
| Conditions flag         | today's WBGT band as a colour                                     | flag         |
| Entry deadline          | next meet's entry cutoff                                          | date         |
| Readiness alerts        | sleep / HRV outside the athlete's band (when connected)           | count        |
| Birthdays this week     | —                                                                 | short list   |

Table 3: Glance widgets: candidates for the four small slots.

| Medium widget (M)       | Shows                                                                                 | Type          |
|-------------------------|---------------------------------------------------------------------------------------|---------------|
| Today's plan (compact)  | the session by group, with print                                                      | text          |
| Conditions strip        | the four windows                                                                      | text + icons  |
| Needs a look (list)     | the names behind the glance count, each linked                                        | list          |
| Most recent session     | who did what against plan, in a paragraph                                             | text          |
| Upcoming meets          | the next three: date, meet, entries due                                               | list          |
| Tasks                   | the organizer                                                                         | list          |
| Team weekly mileage     | eight-week sparkline, planned vs. actual                                              | chart         |
| Pace groups today       | who runs with whom, at what pace                                                      | list          |
| Training load           | acute:chronic ratio per group, as bars                                                | chart         |
| Weather trend           | 48-hour temperature and precipitation line                                            | chart         |
| Recent PRs              | PRs from the last meet                                                                | list          |
| Inbox                   | athlete messages and notes since yesterday                                            | list          |
| Large widget (L)        | Shows                                                                                 | Type          |
| Weekly mileage with CIs | team median weekly volume with an IQR band, planned volume overlaid; per-group toggle | chart         |
| Session review          | the most recent workout, prescribed vs. actual splits per athlete                     | table + chart |
| Week at a glance        | Mon–Sun plan strip, today highlighted                                                 | calendar      |
| Attendance heatmap      | athlete $\times$ day, last four weeks                                                 | heatmap       |
| Readiness grid          | athlete $\times$ (sleep, HRV, load) as traffic lights                                 | grid          |
| Last meet results       | team score, top finishers, PR count                                                   | table         |
| Planned vs. actual pace | one dot per athlete for the last workout, with a CI band                              | chart         |
| Training load timeline  | team ACWR over the season, injuries marked                                            | chart         |
| Next race preview       | course profile and the forecast for the race window                                   | chart + text  |
| Progression             | selected athletes' 5k-equivalent over the season                                      | chart         |

Table 4: Medium and large widgets. Chart widgets are rendered by the product, never assembled by the coach.

resolves per athlete from the model we host, so one line prescribes a different pace to every athlete.

- **Race plans are v2.** When they come they move here from Team board and attach to meet dates: per-athlete goals, predicted pace ranges, eventually MCMC field simulations against expected competition. v1 ships the planner itself.
- **Training paces** (the Pace cards content) surface here as links; the paces themselves live on the athlete page.

**Editing a day in place.** Clicking a day — “Mon 7” — opens that day for editing directly, without leaving the week (Figure 4). The pop-up holds the text entry, the structure the parser made of it, and the **variants by attribute**: the same session prescribed differently to different groups. “5 tempo” for the boys and “4 tempo” for the girls is one workout with two variants, not two workouts, and the coach writes it once. v1 keys variants on the attributes the roster reliably has — gender and pod; a varsity / JV split, or any other tag a program wants, is a *custom attribute*, v2 unless the roster data already carries it, so the first release stays focused on one team. Pods for the day are proposed from the pod settings: the coach can force the number of pods, the number of athletes per pod, and whether pods are mixed- or single-gender; the product fills them from training paces.

Mon, Sep 7 — edit day

2 mi wu, 5 tempo, 3 cd

parsed: wu 2 mi · tempo 5 mi @ T pace · cd 3 mi (10 mi)

variants by attribute

| group | prescription          | Pods    | athletes |
|-------|-----------------------|---------|----------|
| boys  | 2 wu · 5 tempo · 3 cd | A, B, C | 21       |
| girls | 2 wu · 4 tempo · 3 cd | D, E    | 13       |

+ custom split (e.g. — v2 JV)

Pods · proposed from training paces

Pods: auto ✓

per pod: ~6 ✓

single-gender ✓

Save

Preview athlete view

Push to athletes

one workout, four variants

Figure 4: Editing a day in place. One text entry, one parse, and a variant per attribute group (custom splits such as varsity / JV are v2); pods are proposed from training paces under the coach’s pod settings (count, size, gender mixing).

**Views of the same day.** The grid-by-group view above is one way to look at a day. A coach asked for another: a spreadsheet-style sheet of every athlete running that day — one row per athlete with their group and pace — where dragging a row moves the athlete between pods. The automation put Peter in pod B; the coach thinks he should run the faster pace in pod A, and fixes it in one motion. Day and week should support multiple views of the same underlying plan — grid by group, sheet by athlete, the printable list, the athlete-day preview — the way Week wall becomes a view of the week rather than a page of its own.

**Q.** Which views of a day would you actually use — the grid by group, a drag-to-reassign athlete sheet, a printable list? And what is your first move when the proposed pods are wrong?

Build order: the four-zoom shell around the existing planner first; the text parser and the templating layer next; in-place day editing with attribute variants and the athlete-sheet view; race plans in v2.

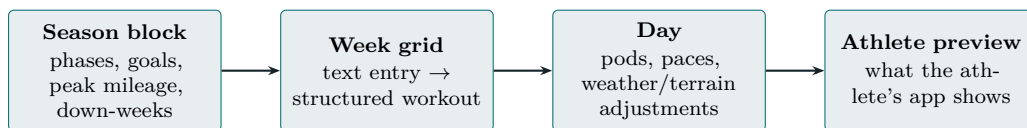


Figure 5: Upcoming: one planning object at four zoom levels. The current Planner / Workout planner / Athlete day split becomes zooming, not separate pages.

Figure 6 is the week zoom as a coach would see it: the zoom selector across the top, a text entry per day with the parsed structure shown beneath it, the race day highlighted, and the block-level context (week of the block, volume target, race plan status) in a rail below.

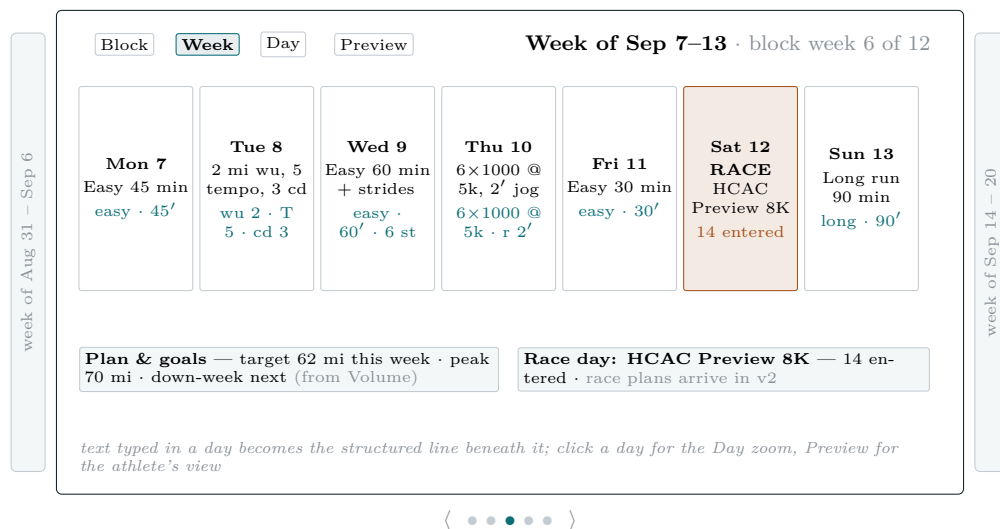


Figure 6: Upcoming, week zoom, shown as a carousel: adjacent weeks peek at the edges and a swipe or arrow moves between them, while the zoom tabs switch level. Each day's free-text entry is parsed into the structured workout shown under it (the load-bearing technical piece); the rail holds the block context. Race plans are v2.

## Team

A team landing page — name, picture, conference/division information, the team document (rules / creeds / values), and the PR board — with five routes (Figure 7): the Roster, the Results repository (import as a pop-up, not a tab), the PR / season-progression boards, Volume (the volume tracking from The Roster tier, plan & goals included), and Operations (money, equipment, the .ics export of the meet and training calendar). Every surface that shows an athlete's name — roster row, result line, PR board entry — opens the same athlete page.

**Q.** Should the athlete page stay reachable only through Team (roster, results, boards), or does it deserve a top-level tab of its own? And does Volume sit better here or under Upcoming?



**Roster attributes.** The roster carries, per athlete, the attributes the rest of the product keys on. In v1 those are **gender** and **pod** (the training-pace group); a **varsity** / **JV** split — or any other tag a program wants — is a *custom attribute*, v2 unless the roster data already carries it, so the first release stays focused on one team. Pods are either set by hand or proposed: the coach chooses the number of pods, or the number of athletes per pod, and whether pods are mixed- or single-gender, and the product estimates current pods from training paces and recent race marks. Workout variants in Upcoming address these attributes directly (“girls: 4 tempo”), which is why they live on the roster rather than in a settings page.

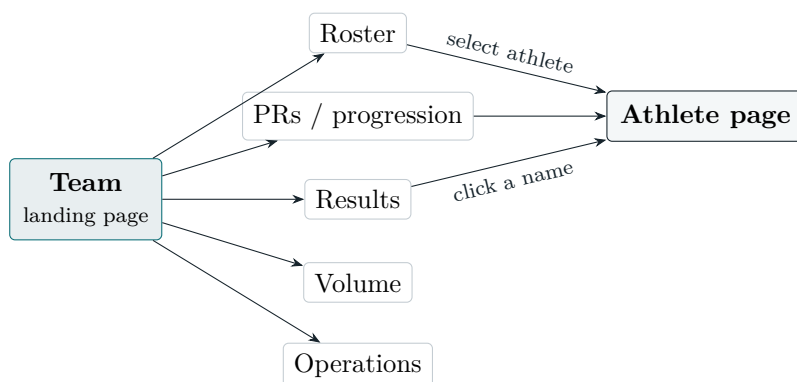


Figure 7: The Team tab. The athlete page is secondary: reached from the roster, the PR / progression boards, or any result line — all three converge on the same object. Volume and Operations stay team-level.

Figure 8 is the landing page with Rose-Hulman Institute of Technology as the example: an identity header (mark, name, location, current season, head coach, conference and division), then a stack of blocks — coaches, athletes, results, PRs, volume, schedule, operations — each of which opens its route. The block order is the coach’s: the handles on the left move a block up or down, so a coach who wants the roster first and the staff list last gets exactly that.



## Rose-Hulman Institute of Technology

Terre Haute, Indiana

**Cross Country · Fall 2026**

Head coach: Derick Lawrence

HCAC · NCAA Division III

↑↓ **Coaches**
collapse ^

| Name            | Title                                                              | Email                    | Phone          |
|-----------------|--------------------------------------------------------------------|--------------------------|----------------|
| Derick Lawrence | Head Men's & Women's Cross Country / Assistant Track & Field Coach | lawrenc1@rose-hulman.edu | (812) 877-8246 |
| Shane Mills     | Head Men's & Women's Track & Field / Assistant Cross Country Coach | mills2@rose-hulman.edu   | (812) 877-8051 |

↑↓ **Athletes** · 77 on the roster · 56 men, 21 women
Roster →

↑↓ **Results** · last: NCAA Division III Championships, Nov 22, 2025
Results →

↑↓ **PRs / progression** · record board, season progression
Boards →

↑↓ **Schedule** · next meet, entries due, .ics
Schedule →

↑↓ **Operations** · money, equipment, travel
Operations →

↑↓ **Volume** · 7-day / 28-day volume, plan & goals
Volume →

Figure 8: The Team landing page, Rose-Hulman as the example. The Coaches block is shown expanded; the rest are collapsed rows that open their route. The ↑↓ handles reorder blocks. Roster counts are the 2025 XC season as held in the data platform.

## The athlete page (three tiers, top to bottom).

### Identity (top, glanceable).

Name, gender, class, hometown, MileSplit and Athletic.net links, and PRs — the public-roster feel the audit of the current Athlete page calls for.

### Secondary.

Recent race results, recent training (a week-over-week mileage graph), attendance / training log, past and current injuries, upcoming registered events (date, name, event), training paces, and stated goals. Person-file material (dietary restrictions, class schedules, paperwork) stays populated further down; anything duplicated elsewhere (e.g. 1-on-1 notes) becomes a link, not a repeated widget.

### Advanced analytics (bottom — v2).

The learned attribute models, deferred to v2 while the data platform matures:

- **Hill Resilience** — is the runner better on flat or hilly courses? Built from course-elevation learning, MileSplit course rankings, FloRatings A/B/C/D, TullyRunners speed ratings, joshm.dev's Crystal Springs Pace Factor, and the golf-slope-rating analog.
- **Weather Impact** — how much adverse weather degrades this runner. The constraint: exact per-race time is needed, and getting it at scale from Athletic.net is not viable (see §2).
- Additional analytical models as desired.

Figure 9 is the page for a real athlete — Billy Allen, a Rose-Hulman sophomore — built from what the platform already holds: headshot and bio from the athletics site, collegiate results from TFRRS, the high-school career from MileSplit (his profile found by name and class year), and the course-adjusted numbers from the ALS fit of §2. The MileSplit and Athletic.net links are badges in the sites’ own marks so a coach recognizes them at a glance.

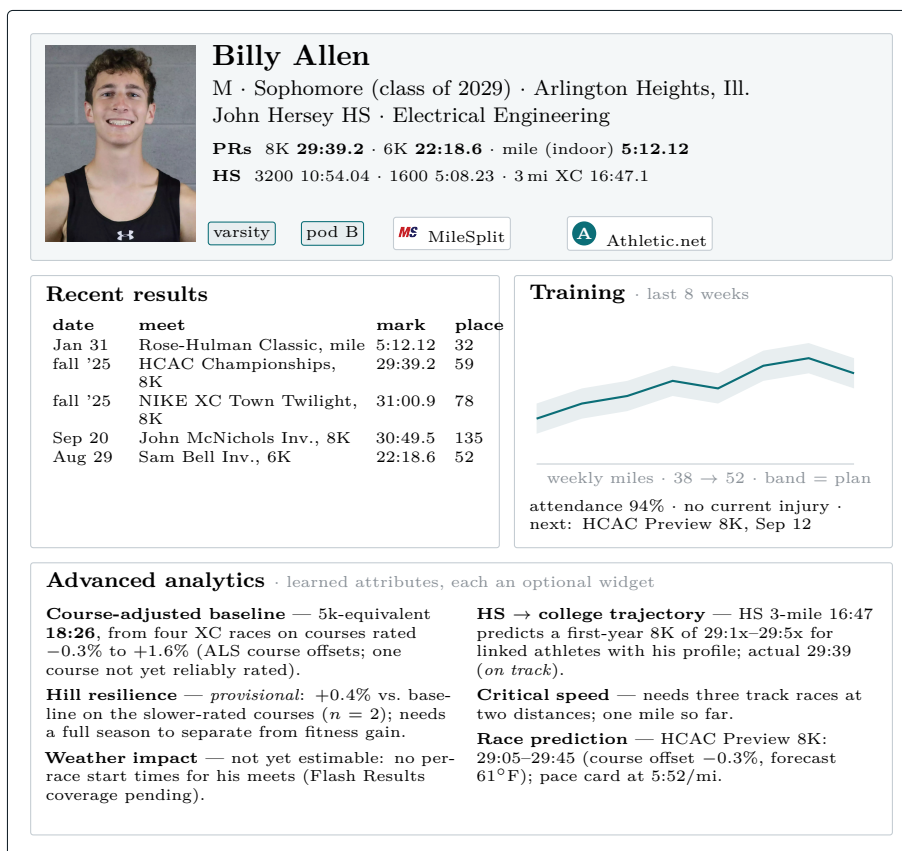


Figure 9: The athlete page for Billy Allen (Rose-Hulman, class of 2029): identity tier with PRs and the HS career via the MileSplit link; recent results and training in the secondary tier; the learned attributes at the bottom, each stating what it needs when it cannot yet be estimated. Bio, PRs and results are real (two meet dates are absent from the TFRRS pages themselves); the training card, attribute chips, and analytics values are illustrative of the output format.

**Analytics that could be added (v2).** Beyond hill resilience and weather impact: a **race-prediction band** per upcoming meet from the baseline, course offset and forecast; **critical speed** / **D'** from track marks; **championship response** (does the athlete run above baseline at championships); **negative-split tendency** from split data where timing provides it; **training response** — the baseline’s movement against mileage and workout load, which is the first place the GetFast training data joins the results data; **consistency** (race-to-race variance, which is itself a selection signal); and the **HS → college trajectory** described in §2, which places a first-year athlete against the linked population with the same high-school profile.

Build order: the identity and secondary tiers ship first — they display data already held. The analytics tier lands as the platform models mature (§2), each model appearing as an optional widget rather than a blank promise.

Recruiting (v2)

Reserved as a tab but not built in v1; recruit-visit mode (masking athlete names) ships in v1 under Settings. When v2 comes, the tab inherits the recruiting file from the current person file, and the HS ↔ college linkage of §2 is its natural engine: a coach looks up a prospect’s verified HS results and course-adjusted marks rather than a self-reported PR.

Settings

One standardized menu (Figure 10) replaces the Housekeeping tier: data importation from Data health, the Setup panels split into their own tabs, the Workout planner’s “Rules baked in,” the Parameters and Predictions pages, feature toggles (the grand-strategy layering — run the product as anything from roster-plus-planner up to the full analytics surface), recruit-visit mode, and bulk import/export, with “How this works” as the help entry. If the paperwork tracked in Data health is medical clearances, it moves to the Team tab’s roster as a who-is-missing-paperwork widget — it is compliance, not data flow. This tab is a mechanical regrouping with no new functionality, which makes it a good early milestone.

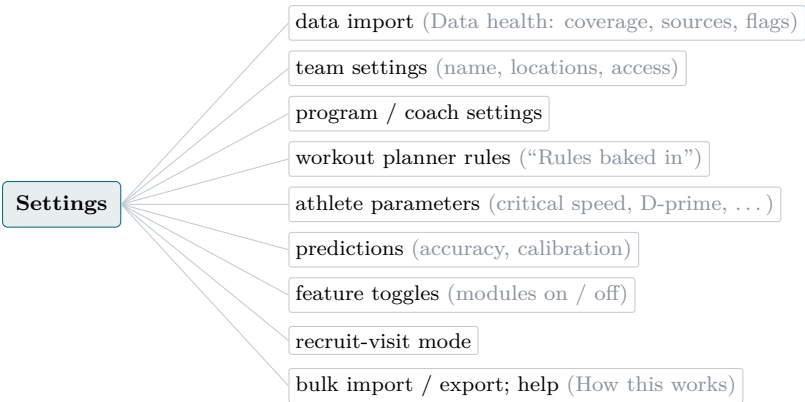


Figure 10: The Settings menu, centralizing what is currently scattered across Housekeeping, the Workout planner, and the athlete sub-pages.

1.6 Wording

Much of the copy reads as too cute, too artsy, or too AI-generated, and should be plainer. Examples:

| Current                                                                                       | Plainer                               |
|-----------------------------------------------------------------------------------------------|---------------------------------------|
| “One line from an athlete at 3:47 is worth more than finding out at 4:10 in the parking lot.” | Cut, or state the actual point.       |
| “Where your attention goes” — “ranked, and every name is a door.”                             | “Needs a look,” with names that link. |
| “Yesterday, rolled up”                                                                        | “Yesterday, summarized.”              |

2 Data and Enrichment Platform

The athlete attributes in §1 (hill resilience, weather impact, and the models to come) are only as good as the data feeding them. This section lays out how that data is sourced, enriched, and linked — and why the platform must **divest fully from Athletic.net** as a scale dependency.

## 2.1 (1) Roster, meet, and team information

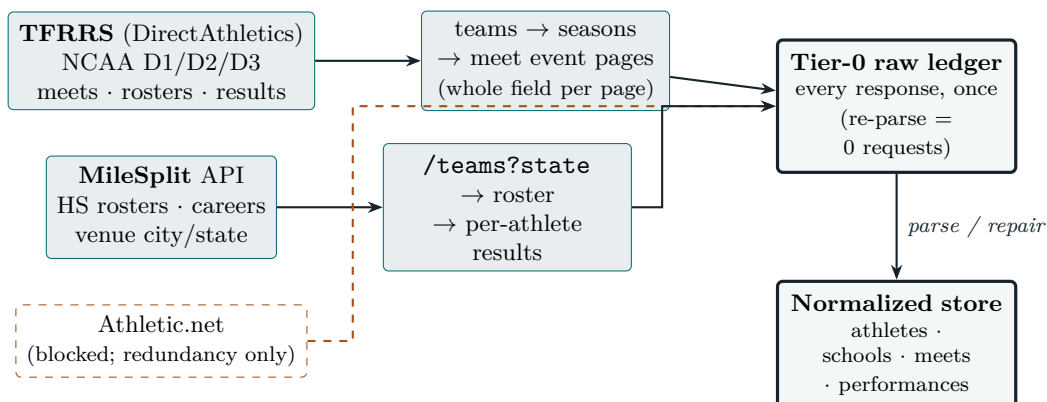


Figure 11: Roster/meet/team sourcing. Every response tees to a raw ledger before parsing, so a parser fix replays at zero request cost. Athletic.net (dashed) is demoted to redundancy.

### How it works, per source.

- **TFRRS (college).** Seeded from the seven NCAA-division region pages, walk teams → seasons → meets. A single meet *event* page carries the whole field (place, athlete, class year, team, mark), so one fetch covers every school in a race. Corpus: 8.1M rows, all three divisions, XC + indoor + outdoor, 2021–2026.
- **MileSplit (HS).** `/teams?state=xx` then the proven per-team, per-athlete ingest path, smallest state first. Polite by default (rate-limited, ledger tee). Note the enumeration cap: `/teams?state` returns at most 999 rows, so the largest states need a per-county or meet-harvest route for full coverage.
- **Divest path.** Athletic.net has the broadest US high-school coverage and the project was built on it, but it is not a dependency to scale on: Cloudflare returns `cf-mitigated=challenge` on sustained crawling (1–2 req/s ran clean for days; 3 req/s drew 24-hour blocks on consecutive days; exit-node rotation did not help — the block is on the traffic pattern, not the address), there is no API, developer program or partnership channel, and its rows carry no race times and collapse relays to one athlete per squad. So: new states come from MileSplit; the existing Athletic.net captures (CA/TX/FL/NY) are retained and merged as a redundancy check; no new pipeline depends on that host.

## 2.2 (2) Course weather and hill information

**Wind is an essential characteristic of a sprint mark.** A 10.90 at +3.0 and a 10.90 at −1.0 are different performances, and the rule book agrees (marks over +2.0 m/s are not record-legal). For the wind-legal events — 100, 200, the hurdles, long and triple jump — the measured gauge reading is stored on the result row itself (Athletic.net carries it on ~9% of track rows; TFRRS event pages carry a WIND column), and every sprint analytic is conditioned on it. Forecast wind from the weather model is the fallback for distance events, never a substitute for the gauge on a sprint.

**Hill / course difficulty** (three complementary signals):

1. **Empirical ALS course offsets (built).** A two-way log-time model —  $\log(\text{equiv5k}) = \text{ability} + \text{offset}$ , i.e. an athlete-season term plus a course term. **ALS** is *alternating least squares*: hold the

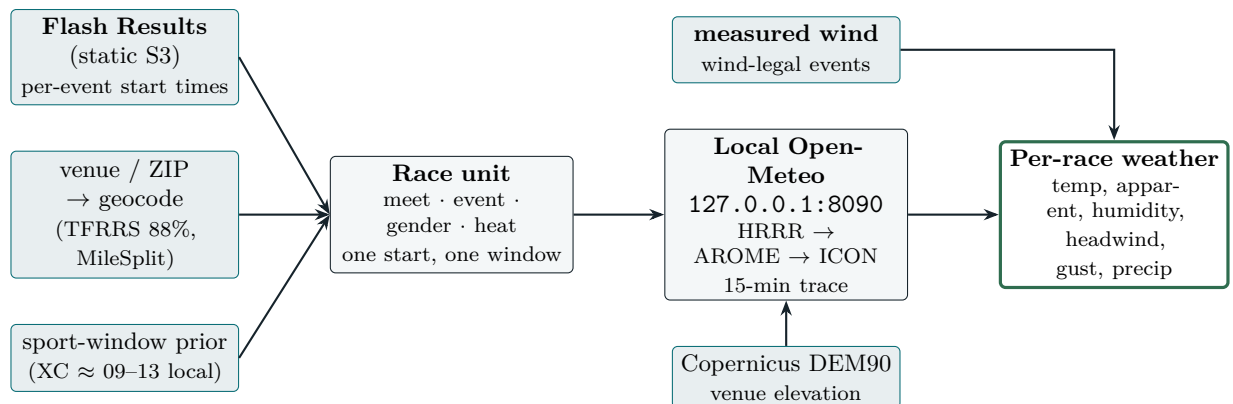


Figure 12: Weather + elevation per *race*, not per meet. The 11 am women’s JV and the 3 pm men’s varsity get different windows. Wind for sprints comes measured from the results themselves.

course offsets fixed and solve each athlete’s ability as the least-squares (mean) fit to their races; then hold abilities fixed and solve each course’s offset the same way; alternate until nothing moves (400 rounds), with the offsets anchored to mean zero so the split is identified. It needs only athletes who race more than one course. Validated split-half  $r = 0.98$ ; recovered Crystal Springs (+72s) and Woodbridge (−50s) with no external input. It measures what terrain *costs*, but bundles hills with footing and weather.

2. **External course ratings (cross-check + decomposition).** TullyRunners speed ratings, MileSplit course rankings, FloRatings A/B/C/D, joshm.dev Crystal Springs Pace Factor, golf-slope analog. Joining these and correlating with the ALS offset validates our number and separates terrain-driven from conditions-driven difficulty.
3. **True elevation gain (targeted).** GPS/GPX course traces through the DEM give cumulative climb per km — the only physical hill metric, but per-course external data. Reserve for marquee courses.

**Held-out test on 100 athletes per division.** To check that the offsets carry information rather than fit noise, 100 athlete-seasons with at least three XC races were drawn at random from each of D1, D2 and D3 (seed 42), one race per athlete was held out, and the model was re-fit on the remaining 619,056 rows (151,239 athlete-seasons, 3,410 meets). For each held-out race the question is whether the *course offset* explains what the athlete’s own baseline cannot (Figure 13, Table 5): the correlation between the athlete’s deviation from baseline and the course’s offset is 0.50–0.76, the prediction error falls 20–36% once the offset is applied, and a permutation null (offsets shuffled across the held-out races, 2,000 draws) never reaches the observed correlation ( $p < 0.0005$ , null s.d. 0.10). D2 is the weakest, consistent with its thinner meet overlap.

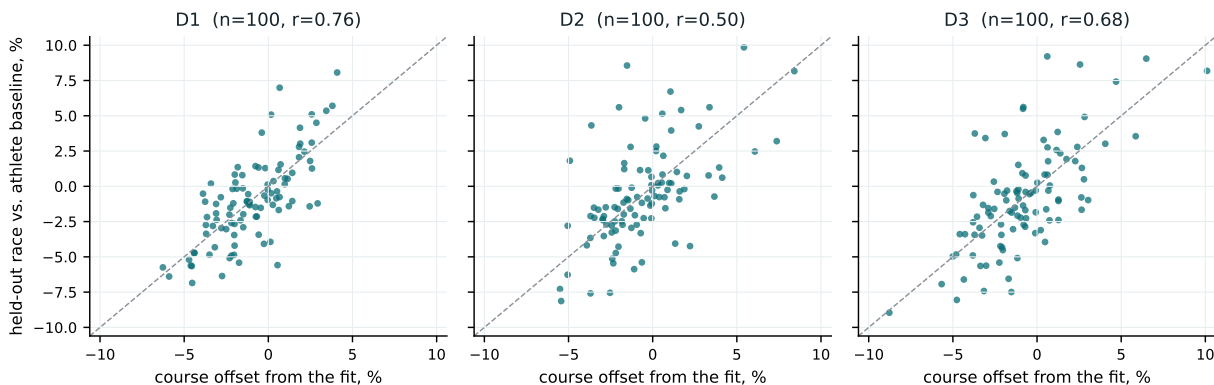


Figure 13: Held-out race vs. course offset, 100 random athletes per division. Each point is one held-out race:  $x$  is the course’s fitted offset,  $y$  is how far the athlete ran from their own baseline that day. Points on the dashed line are fully explained by the course.

|    | $n$ | $r(\text{dev.}, \text{offset})$ | $r$ pred. | $r$ base | MAE base | MAE w/ offset |
|----|-----|---------------------------------|-----------|----------|----------|---------------|
| D1 | 100 | 0.76                            | 0.98      | 0.95     | 28.4 s   | 18.2 s (−36%) |
| D2 | 100 | 0.50                            | 0.97      | 0.97     | 35.4 s   | 28.3 s (−20%) |
| D3 | 100 | 0.68                            | 0.98      | 0.97     | 35.3 s   | 24.1 s (−32%) |

Table 5: Held-out validation of the ALS course offsets. MAE is on the 5k-equivalent time. Permutation  $p < 0.0005$  in every division.

Two caveats found while running it. The fit was made on the corpus as it stands, which still holds 2,705 duplicate XC rows across 46 meets from the July legacy sync (607 with a conflicting distance label) — small, and a targeted delete, but one affected meet’s offset is wrong until it is done. And 259k XC rows belong to schools whose TFRRS slug did not resolve to a division; the samples are drawn from the 405k that did.

**Assessment.** Once weather is attached per race, re-fit the ALS *net of the weather term*: the course offset then stops absorbing that day’s heat and becomes a terrain-only score — the cleanest hill metric derivable without a GPS trace.

## 2.3 (3) Linking college and high-school runners

There is no crosswalk: zero athletes carry both an Athletic.net id and a TFRRS id, and MileSplit’s `tfrsId` field is null on every bio examined. Linkage is therefore *probabilistic*, and the honest deliverable is the ceiling.

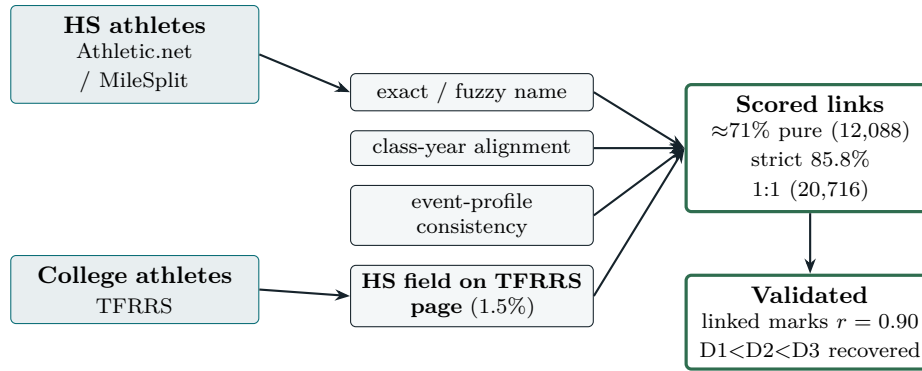


Figure 14: Probabilistic HS→college linkage. No exact key exists; a scored combination of name, year and event profile reaches  $\sim 71\%$  precision, and TFRRS athlete pages carry a literal High School field (1.5% of pages) that opens a path to a *real* crosswalk by crawling those pages directly.

**Why link at all.** The link is what turns two separate result histories into a *trajectory*. With it the platform can say how athletes with a given high-school profile developed in college — expected first-year 8K from a 3-mile PR, typical improvement by event and by division, who bloomed late — and place any current athlete against that population: Billy Allen’s 16:47 3-mile became a 29:39 8K as a freshman, and the linked population says whether that is ahead of, on, or behind the curve. That **potential-growth** view is the core of recruiting (v2): a prospect’s verified HS marks projected forward, rather than a self-reported PR read at face value. It also benchmarks a program (what HS profile did our athletes who succeeded arrive with) and lets course-adjusted HS and college marks be compared on one scale.

**The route to a real crosswalk.** TFRRS athlete pages carry a literal “*High School:*” field on 1.5% of archived pages (roster pages do not). Crawling athlete pages directly — rather than inferring identity from names — upgrades the linkage from probabilistic to keyed for the athletes who have it, and is the recommended next step for the recruiting and progression models.

## 2.4 (4) Finding meets to enter

A coach’s recurring question — what is coming up, and when do entries close — has a scalable answer for high school and an open one for college.

- **High school: MileSplit’s meet feed.** The meets endpoint, `/api/v1/meets` with a `state` parameter, returns a rolling window of roughly 60 days per state with `dateStart`, `dateEnd`, `season`, venue city/state and a `registrationActive` flag (probed 2026-08-22: 100 Indiana meets, 65 with registration open). It has no deadline field and ignores `page`, so the pattern is: poll each state once a day (51 requests), diff against what was seen, and fetch the meet’s own page once for any *new* meet with registration open to read the entry deadline. Filter by distance from the team’s geocode, by level, and by season, and the result is an “upcoming meets near you, entries due by” list that costs a few hundred requests a day nationally.
- **College: the team’s own schedule page.** TFRRS lists only past meets, and DirectAthletics — where college entries actually happen — answered our client with a CloudFront 403. But college athletics sites run on a handful of templates (Sidearm, PrestoSports), and Rose-Hulman’s served us without challenge: the schedule page carries every scheduled meet with date, host and an `.ics` export. Reading a team’s schedule from its athletics site is uniform across the template



and is the college route; entry deadlines still come from the coach’s DirectAthletics account or the host’s meet page.

## 2.5 Build order

Steps that cost nothing against any host come first:

1. Venue + geo columns; reparse venue from the archived TFRRS/MileSplit payloads. **Zero requests.**
2. Wind column on performances; backfill from archived Athletic.net payloads (wind-legal events). **Zero requests.**
3. Flash Results time crawler — new host, static S3, its own polite profile and ledger. Start with the NCAA championship/regional set and measure the match rate before scaling.
4. Weather adapter: emit a per-race manifest, run the existing `~/weather` Open-Meteo pipeline unchanged, reduce to race-window aggregates.
5. Course difficulty: ship the ALS offset, join external ratings as a check, then re-fit net of weather for a terrain-only score.

## 2.6 v2 — Recruiting: considerations and goals

Recruiting is deliberately a v2 product: it needs the linkage and the trajectory models above to be real before it is more than a contact list. Goals, in order:

1. **Prospect lookup with verified marks.** A name resolves to the athlete’s MileSplit / Athletic.net record, and the page shows their marks as the platform holds them — course-adjusted, wind-conditioned where it applies — not as reported.
2. **Potential growth.** The HS  $\rightarrow$  college trajectory placed against the linked population: expected first-year marks, the range, and the comparable athletes it is drawn from, by division.
3. **Fit to program.** Where the prospect would sit on this roster (varsity / JV line, pod) and against the conference, using the same projections as the Team tab.
4. **Watch lists and timelines.** Prospects by class year with contact-period dates, the recruiting file that today lives in the athlete person file, and notes.
5. **Recruit-visit mode** stays in v1 under Settings: masking athlete names on a campus visit is a display toggle, not a recruiting feature.

Considerations: NCAA contact rules vary by division and must be encoded, not assumed; high-school data comes from sources with redistribution terms (MileSplit’s non-commercial etiquette, Athletic.net’s ToS), so a commercial recruiting product likely needs a licensing conversation before it ships; and the linkage is probabilistic ( $\sim 71\%$  pure today), so every projection must carry its confidence and its comparables, never a bare number.